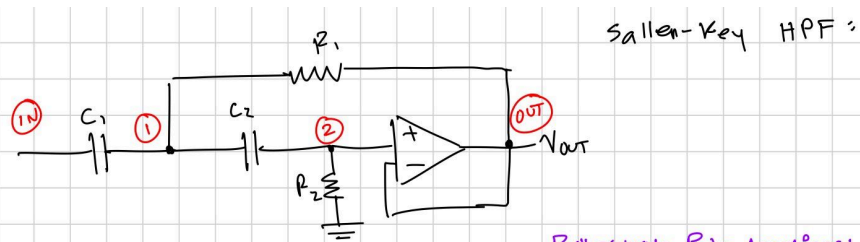


Design Project: Part 2

NO AI USED IN ASSIGNMENT



Butterworth Pole locations:

2nd order 0.7071 0.7071

$$s = -0.7071 \pm j0.7071$$

$$H(s) = \frac{s^2 R_1 R_2 C_1 C_2}{s^2 \underbrace{R_1 R_2 C_1 C_2}_a + \underbrace{R_1 (C_1 + C_2)}_b s + 1}$$

$$a = R_1 R_2 C_1 C_2$$

$$b = R_1 (C_1 + C_2)$$

$$(s^2 + 0.7071 - j0.7071)(s^2 + 0.7071 + j0.7071) = s^2 + s + 1$$

$$s^2 + 1.4142s + 1$$

$$\therefore H(s) = \frac{1}{s^2 + s + 1}$$

Assume desired cutoff frequency is 1000 Hz, $\omega_c = 1000 \frac{\text{rad}}{\text{sec}}$

$$C_1 = C_2 = 1 \text{ F}$$

$$a = R_1 R_2 C_1 C_2 = 1$$

$$b = R_1 (C_1 + C_2) = 1.4142$$

$$R_1 = 0.7171$$

scaling $R'_1 = k_m R_1$

$$k_m = 10,000$$

$$R'_1 = 7171 \Omega$$

$$R_2 = \frac{1}{R_1}$$

$$R'_2 = k_m R_2$$

$$R''_2 = R'_2; R''_1 = R'_1$$

$$R_2 = 1.4142$$

$$R'_2 = 14142 \Omega$$

$$C'_1 = \frac{C_1}{1000} = 1 \text{ mF} = C'_2; C''_1 = \frac{C'_1}{\omega_c} = 1 \mu\text{F} = C''_2$$

Figure 1: Solving for Sallen-Key Impedance Values

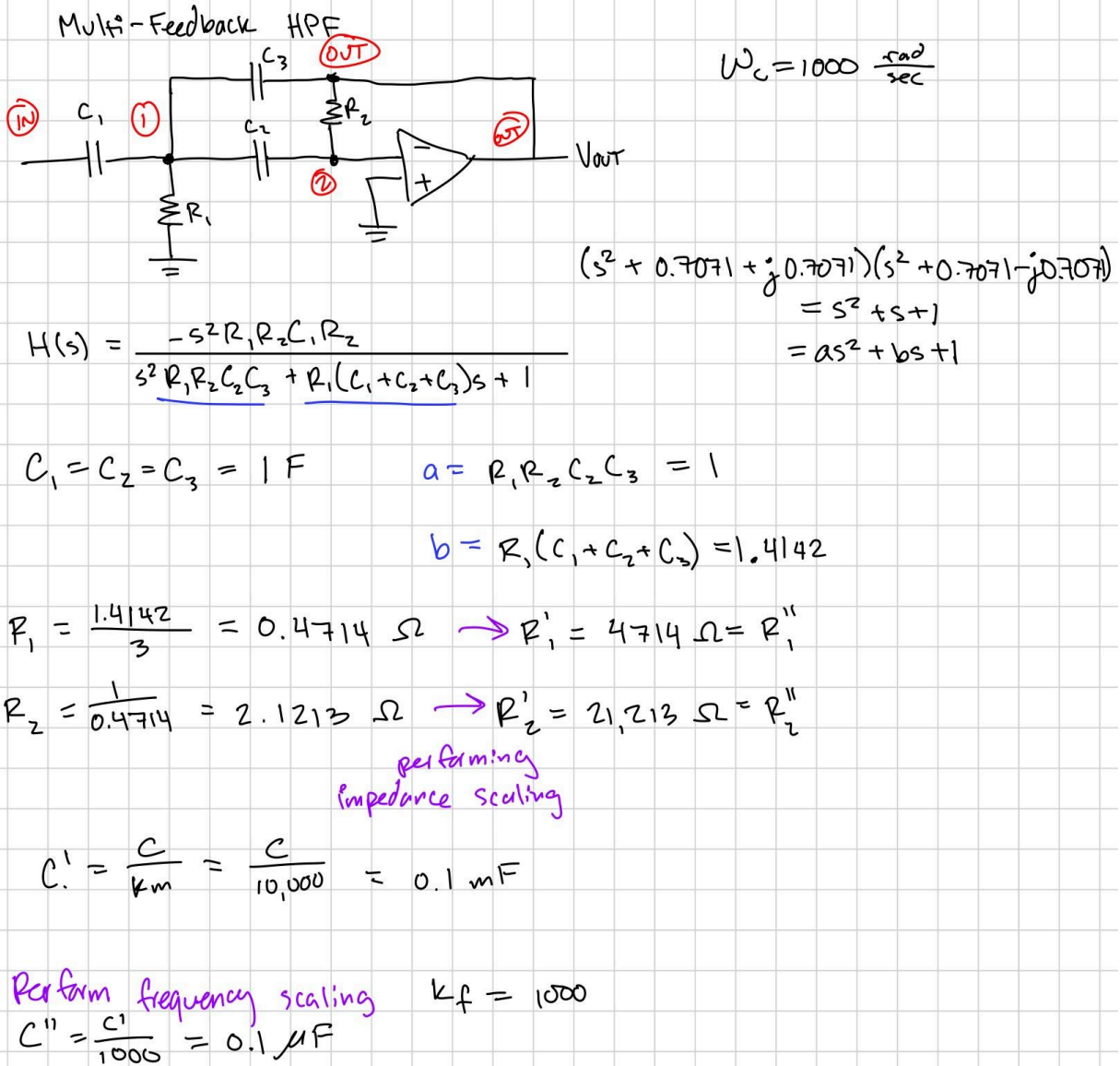


Figure 2: Solving for MultiFeedback Impedance Values

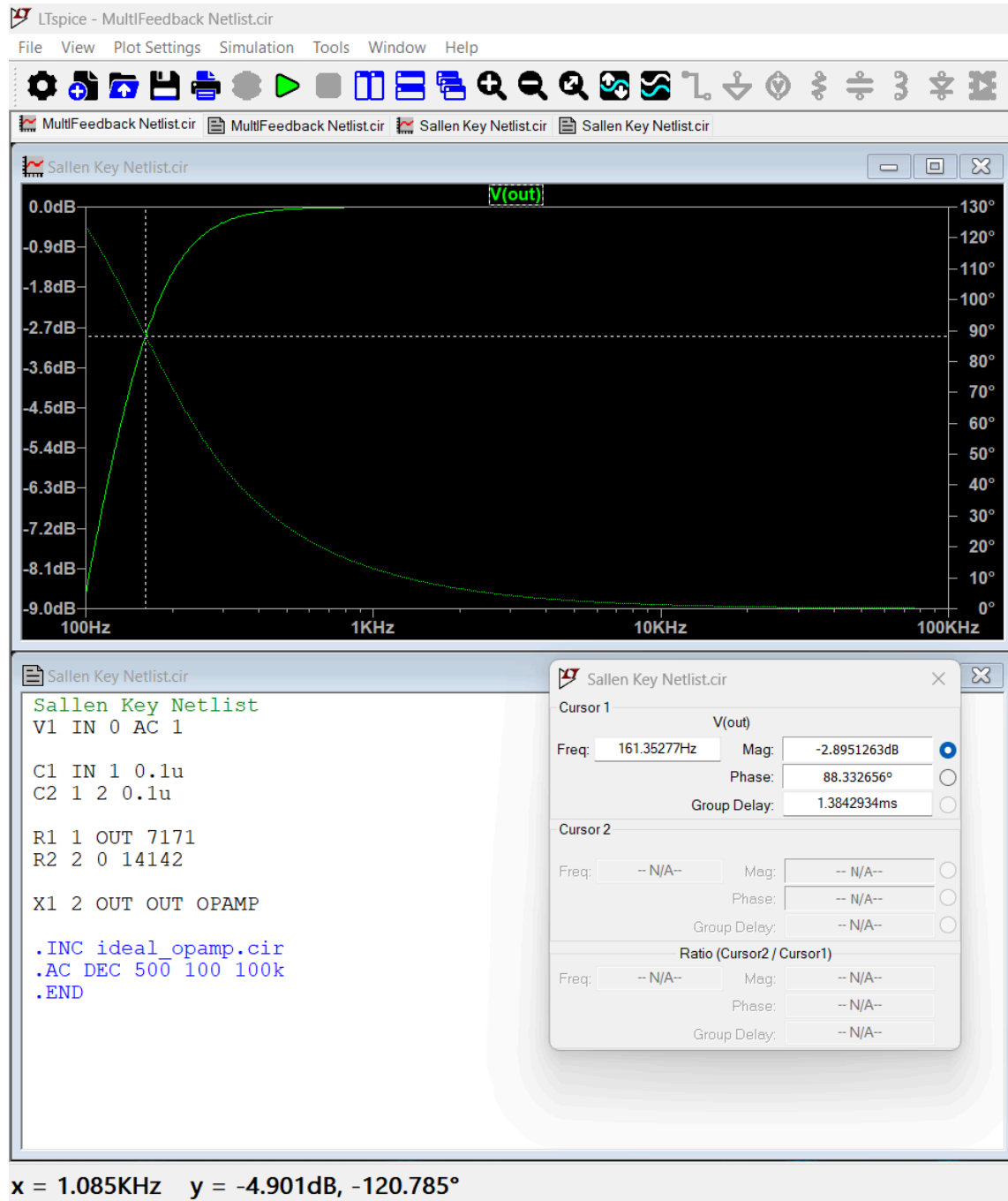


Figure 3: Sallen-Key Netlist and Plot

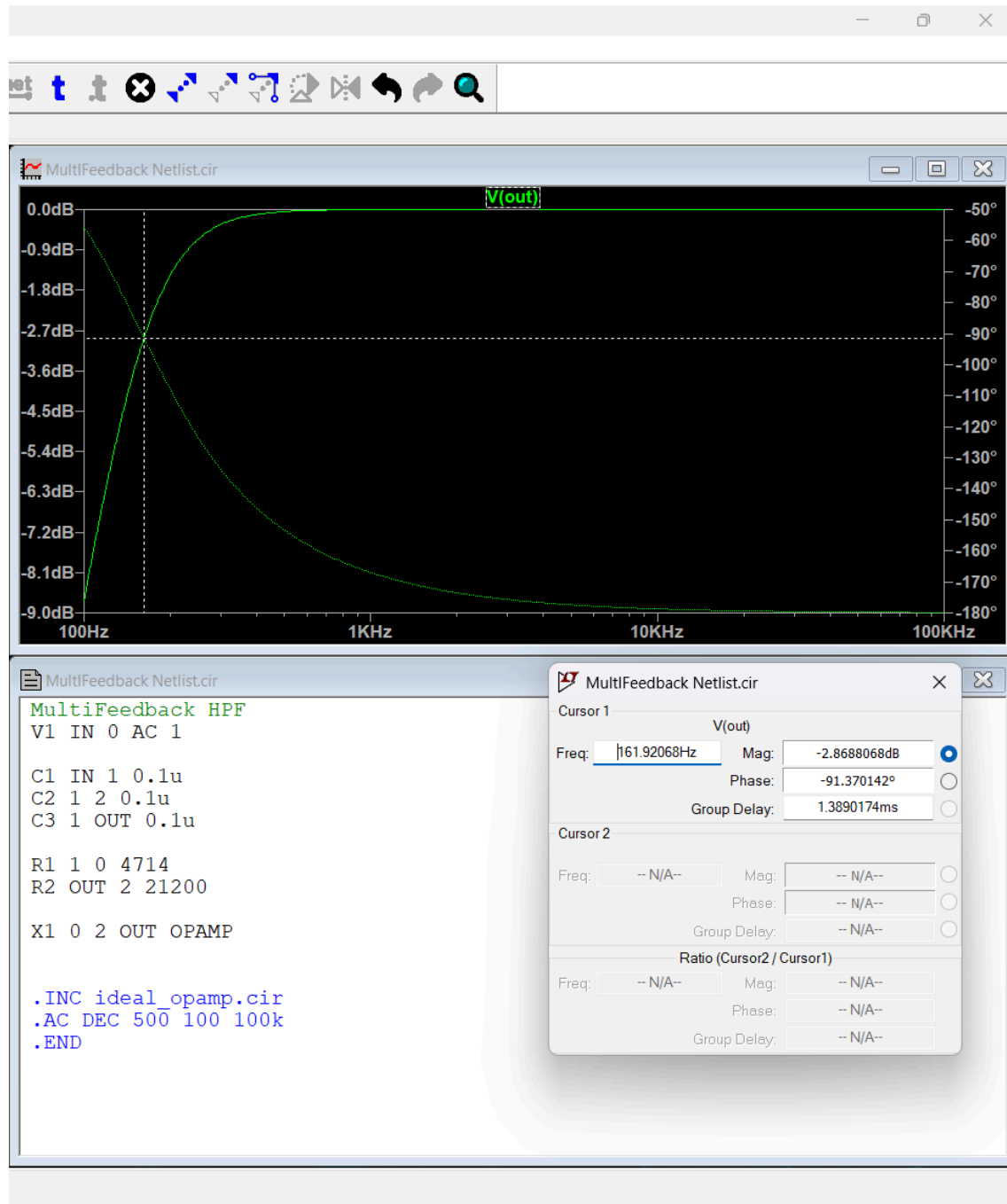


Figure 4: MultiFeedback Netlist and Plot

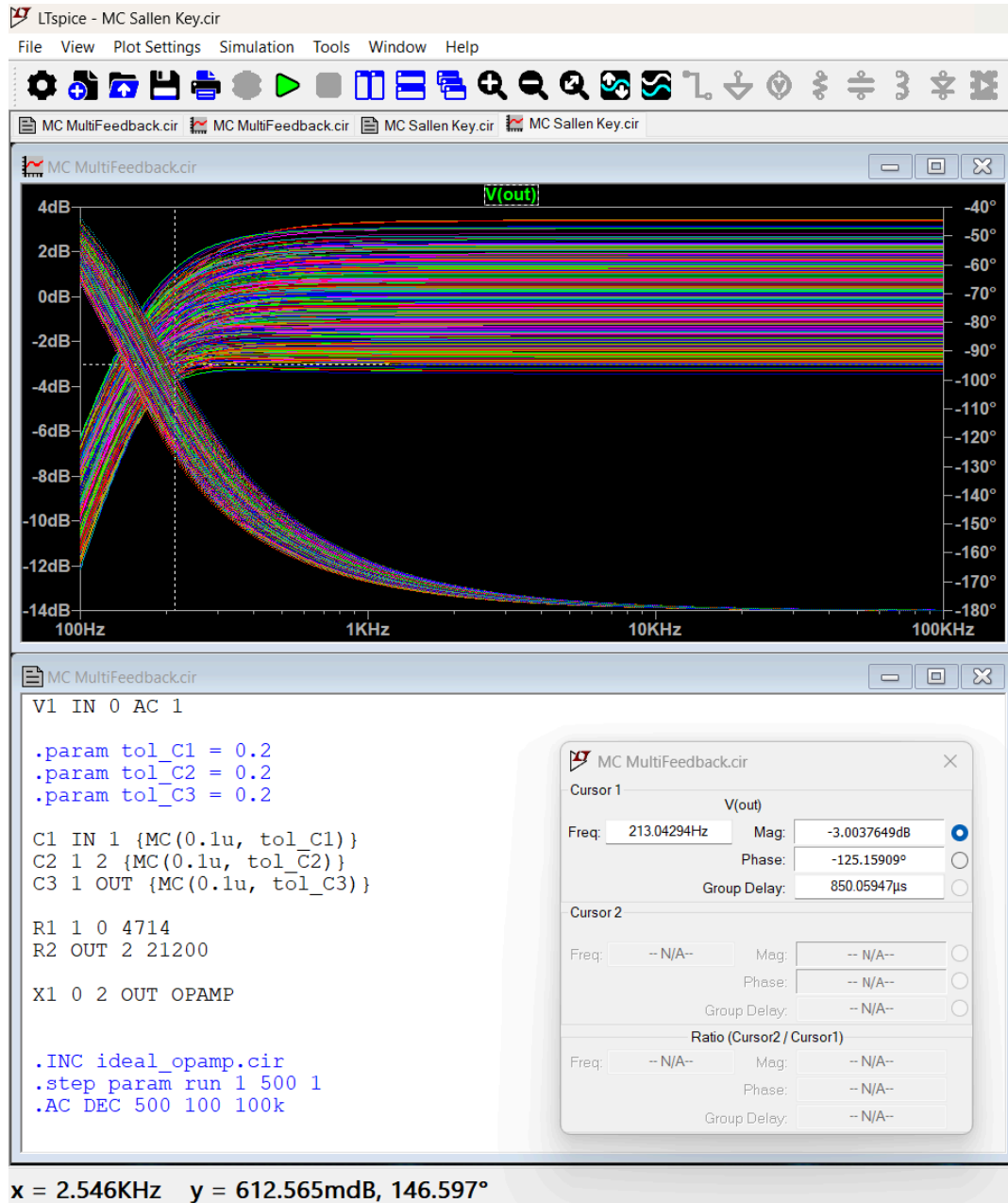


Figure 5: Monte Carlo MultiFeedback Netlist and Plot

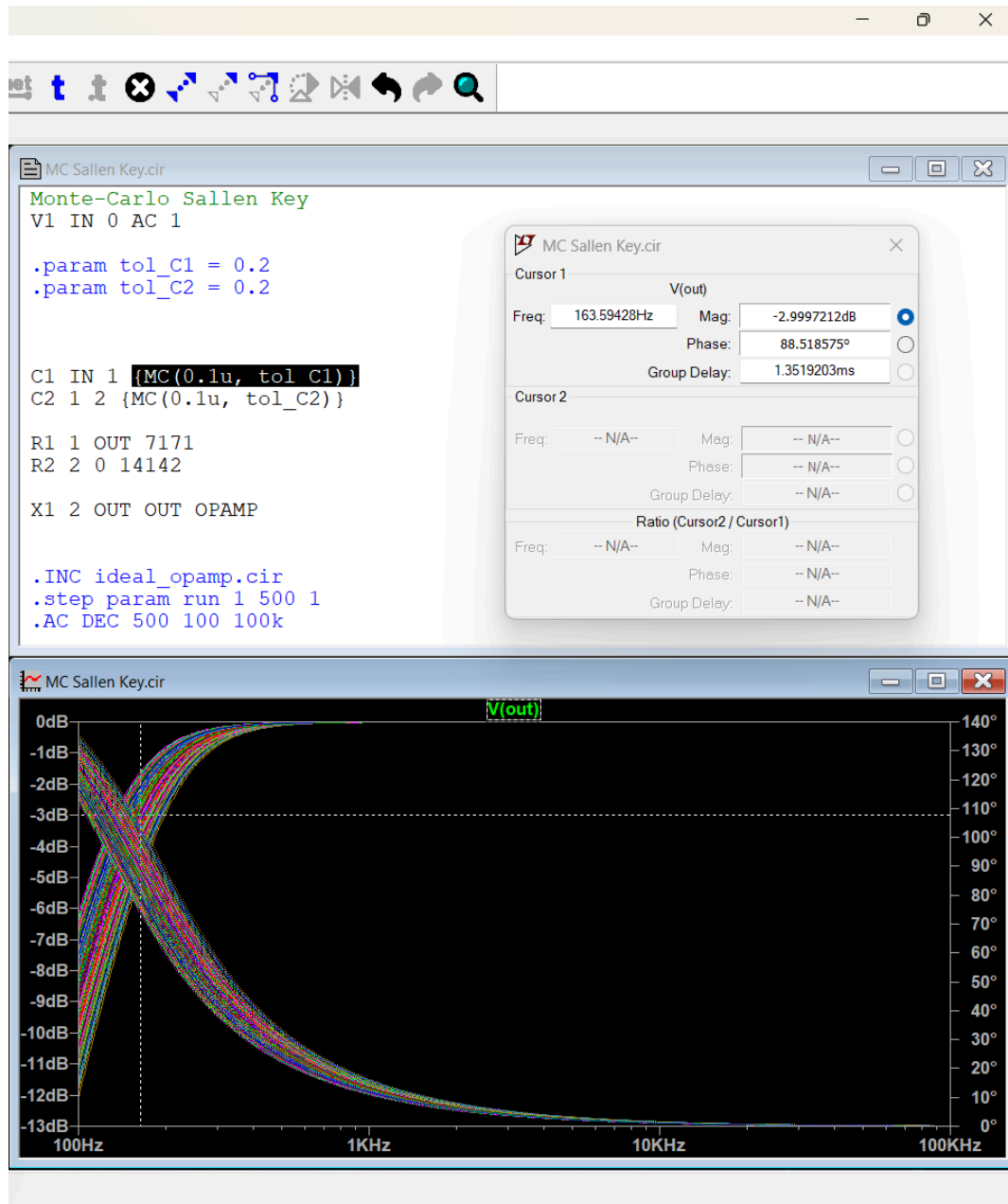


Figure 6: Monte Carlo Sallen Key Netlist and Plot

6. Based on the results from the Monte Carlo simulations, the Sallen-Key seems to be the preferred circuit structure. At -3dB of the Monte Carlo Sallen-Key, the frequency is at 163 Hz, which is about 1024 rad/sec, still close to the 1000 rad/sec cutoff frequency. When the Monte Carlo MultiFeedback HPF is at -3dB, the frequency is at about 1338 rad/sec, which is well over the cutoff frequency that we determined.